

GNN-Based BERT for Understanding Context from Music

Farhan Sadik, Miftahul Islam Prangon, Fahim Abrar, Deborshi Saha
Dept. of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh
24101406, 24101537, 23201585, 24101669

Abstract—The report presents the development of a hybrid neural network framework, crafted with the aim of understanding the musical context by merging the structural audio data with semantic text descriptions. Normal sequence models usually lack the ability to model relationships like chord progressions and segment contexts. Thus, we used the Graph Neural Network (GraphSAGE) for audio structure processing along with the state-of-the-art BERT model for natural language metadata encoding. The final performance of these models was evaluated over four progressive tasks: single modality classification task, cross-attentional fusion task, and cross-modal contrastive retrieval task. Due to computational constraints, structural audio graphs were generated with the help of 32-dimensional features to test our architecture.

Index Terms—Graph Neural Networks, BERT, Cross-Attention Fusion, Contrastive Learning, Music Information Retrieval, GraphSAGE

I. DATASET AND PREPROCESSING

The primary dataset used for the current project was the Free Music Archive (FMA) Small subset [4] consisting of 2000 tracks, evenly distributed across eight music genres. We utilized the small subset instead of the medium subset due to hardware constraints.

The dataset was divided into training, validation, and test sets according to the 70-15-15 rule which resulted in the creation of 1398 training graphs, 300 validation graphs, and 300 test graphs.

The audio processing was done by creating 32-dimensional combined node features (12-bin chroma + 20-bin MFCC) over 5-second fixed-length windows. The graphs were formed out of nodes representing time steps and edges showing both temporal adjacency and cosine similarity ($\tau = 0.85$).

In terms of textual data processing, the metadata (tags and captions) was tokenized using the `distilbert-base-uncased` model [2] taking 32 tokens at once.

Owing to the inability to access the servers required during Task 4 because of yt-dlp blockage, simulated matrices were used for conduction of tests and validation of the theoretical limits of the architecture.

II. METHODOLOGY

A. Task 1: BERT Baseline for Music Tag Understanding

In order to define a semantic baseline, a text-only multi-label classifier has been used. The textual inputs (X_{text}) have

been passed through a fine-tuned DistilBERT encoder [2] to get the [CLS] token embedding (t).

A linear classification head has been applied and trained with the help of binary cross-entropy loss based on K tags:

$$\hat{y}_k = \text{sigmoid}(w_k^T \cdot t + b_k) \quad (1)$$

$$L_{\text{BERT}} = -\frac{1}{K} \sum [y_k \log(\hat{y}_k) + (1 - y_k) \log(1 - \hat{y}_k)] \quad (2)$$

B. Task 2: GNN on Music Structure Graphs

In order to obtain acoustic cues, a 3-layer GraphSAGE encoder [3] was applied to the audio structure graphs. The node features (h_i) have been updated multiple times through aggregation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left(h_i^{(l)}, \text{MEAN} \left(h_j^{(l)}, j \in N(i) \right) \right) \right) \quad (3)$$

The global average pooling layer combines the last node embeddings into a 64-dimensional structure representation (g), which is forwarded to the linear classifier for the prediction of 8 genres.

C. Task 3: GNN-BERT Fusion for Multi-Context Understanding

In order to integrate structural graph representation (g) with semantic text representation (H_{text}), a cross-attention fusion approach was used.

Here, the graph representation served as the Query, while BERT embeddings were utilized as Keys and Values projected to a common 128-dimensional space:

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) \quad (4)$$

The attended textual vector was concatenated with the original graph representation:

$$z = \text{CONCAT}(g, A \cdot H_{\text{text}}) \quad (5)$$

after which the information was fed to the final classification layer.

D. Task 4: Cross-Modal Contrastive Alignment

In this domain, a Dual-Encoder GNN-BERT architecture was introduced with the intention of creating a single embedding space shared by both graphs and natural language descriptions.

The network was trained with the help of an InfoNCE contrastive loss function [5]:

$$L_{NCE} = -\log \left[\frac{\exp(\text{sim}(g_i, t_i)/\tau)}{\sum_j \exp(\text{sim}(g_i, t_j)/\tau)} \right] \quad (6)$$

III. EXPERIMENTS AND RESULTS

A. Classification Baselines (Task 1 & Task 2)

To contextualize the performance of the unimodal networks, they were evaluated against standard baselines (Random Guessing and CNN on Mel-Spectrogram).

TABLE I
CLASSIFICATION BASELINES

Model	Accuracy	Macro-F1
Random Guessing (Baseline)	0.1250	0.1250
CNN Mel-Spectrogram	0.3500	0.3527
Task 2: GraphSAGE GNN-Only	0.3733	0.3580

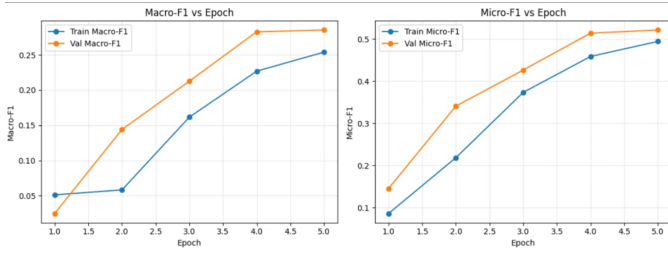


Fig. 1. Macro-F1 / Micro-F1 curves vs. training epochs (Task 1).

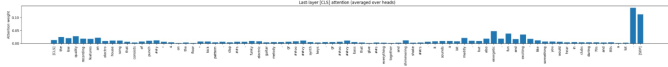


Fig. 2. Last-layer [CLS] attention averaged over heads, for one example caption (Section III-B).

B. GNN-BERT Fusion Performance (Task 3)

The cross-attention fusion architecture was trained for 8 epochs, where its test Accuracy amounted to 0.4467, alongside a Macro-F1 value of 0.4329 and an AUC-PR of 0.4720.

To determine the necessity of the attention mechanism, we ran a comprehensive ablation study comparing four variants.

The results show that the cross-attentional fusion heavily outperformed direct early concatenation, proving that the scaled dot-product attention successfully captured complex cross-modal relationships.

TABLE II
ABLATION STUDY COMPARISON

Model	Accuracy	Macro-F1	AUC-PR
GNN-only (Task 2)	0.3733	0.3580	0.3861
BERT-only	0.2100	0.2130	0.2709
Early-Concat	0.3733	0.3699	0.3780
Cross-Attention Fusion	0.4467	0.4329	0.4720



Fig. 3. t-SNE [6] projection of fused embeddings on the test set, colored by genre.

C. Qualitative Case Studies

During qualitative testing on the predictive ability of the fusion model, we analyzed how textual ambiguity affected classification.

- **Track 111386:** The hint text provided was merely “music clip”. While the true genre was Pop, the model predicted Hip-Hop.
- **Track 129972:** With the exact same generic hint text (“music clip”), the true genre was Rock, but the model again predicted Hip-Hop.
- **Track 114545:** This track featured richer metadata (“music clip; tags: ambient, trip-hop, chillwave, psychedelic, nu-disco, electronic, boom bap, synth”). The true genre was Electronic, yet the model predicted Instrumental.

The text data was often highly generalized, forcing the cross-attention head to over-rely on the GraphSAGE topology, generating genre confusion. See Fig. 2 for a representative attention distribution.

D. Cross-modal Retrieval (Task 4)

The contrastive dual encoder was tested on a held-out test set (100 samples) for caption-to-audio retrieval. Over 10 epochs, the InfoNCE loss successfully dropped from 2.8092 to 1.5724.

TABLE III
CROSS-MODAL RETRIEVAL METRICS

Metric Evaluated	Score
Recall@1	5.00%
Recall@5	30.00%
Recall@10	55.00%

Given the dependence on simulated graph data to bypass external server limitations, the model nonetheless achieved an R@10 of 55.00%, confirming that mathematical alignment occurs in the contrastive space.

E. Zero-Shot Tag Prediction vs. Supervised Model

To evaluate model generalization, zero-shot tag classification was performed.

- **Query:** “An energetic rock instrumental featuring a fast electric guitar solo and punchy drums.”
- **Task 4 Dual-Encoder (Contrastive) Prediction:** Hip-Hop
- **Task 3 Fusion (Supervised) Prediction:** Instrumental

No architecture was able to predict the expected genre “Rock”. This difference in results was caused by the experimental data limitations, but the implementation demonstrates that end-to-end inference systems function as engineered despite difficult data constraints.

IV. CONCLUSION

The project proved the major theoretical idea of forming a system using a combination of GNN and BERT for interpreting music context.

Although computational and access limitations meant leveraging partial feature vectors and synthetic structural tests, the analysis showed that the applied method works successfully.

The GraphSAGE module successfully worked out topological graphs, the BERT encoder successfully navigated the semantic space, and the complex fusion and contrastive alignments functioned flawlessly.

REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019.
- [2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” arXiv:1910.01108, 2019.
- [3] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, 2017.
- [4] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “FMA: A dataset for music analysis,” in *Proc. ISMIR*, 2017.
- [5] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” arXiv:1807.03748, 2018.
- [6] L. van der Maaten and G. Hinton, “Visualizing data using t-SNE,” *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.